

Was It Worth It? Reading a Cost-Benefit Result

Module 2 · Day 3, Session 1: The Assumption That Decides the Answer

3ie

0 · Where the week has got to

From “how big” to “worth it”

Day 1 three questions for any number; what a fair comparison is

Day 2 three ways to estimate an effect: DiD, RDD, matching

Today what the effect is *worth*, and which assumption decides it

On Day 2 the saving missed the 1,000 AED rule. **Could the same programme still be worth funding?** Hold your answer; we test it today.

Where the 816 AED comes from

R Code

↺ Start Over

▶ Run Code

```
1 offered <- subset(gw, treatment_neighborhood == 1)
2 m <- tapply(offered$waste_management_costs, list(offered$enrolled, offered$round), mean)
3 round(data.frame(before = m[, "0"], after = m[, "1"], change = m[, "1"] - m[, "0"],
4                 row.names = c("did not enrol", "enrolled")))
```

	before	after	change
did not enrol	2079	2230	151
enrolled	1449	784	-665

From Day 2: which two numbers do you subtract to get the saving?

The two changes: **-665 - 151 = -816 AED** a year per business, with a 95% interval of about 753 to 880. It missed the 1,000 AED rule. Today we ask a different question of the same number.

1 • The headline

The decision, and the ratio

GreenWaste cost the authority **1,800 AED per business** to deliver, once. It saved each business about **816 AED a year**.

A **benefit-cost ratio** compares the two: what you get back, divided by what you spend, with future money counted as worth less than money today.

Cost	1,800 AED per business, one-off
Saving	816 AED per business per year
Decision rule	a ratio above 1 pays for itself
Businesses	2,964 enrolled

The headline number

1.87

Benefit-cost ratio, GreenWaste

AED spent per business **1,800** AED returned per business **3,366** AED net, whole programme **4,641,617**

For every AED spent, about 1.87 AED comes back. This is the programme that missed the Day 2 rule, and both statements are correct.

The rule asks about the saving **in one year**. The ratio adds up the savings **over several years** and sets them against a one-off cost. Different questions can give different answers.

[MENTI PLACEHOLDER] Would you approve it?

On a ratio of **1.87**, what would you do?

Approve · Reject · I need to know something else first

Go to **menti.com**, code **1234 5678**, or scan the code.



What went into that number

Step	Value	Source
Cost per business	1,800 AED	programme records
Annual saving	816 AED	measured, Day 2
Years of saving assumed	5, starting in year 2	assumed
Discount rate assumed	5% a year	assumed
Savings, in today's money	3,366 AED	calculated
Benefit-cost ratio	$3,366 / 1,800 = 1.87$	calculated

The two red rows were not measured: the analyst chose them. **Which of the two would you expect to move the ratio more?**

2 · Build the ratio yourself

A saving next year is worth less than one today

R Code [↺ Start Over](#)

[▶ Run Code](#)

```
1 round(savings_table(rate = 5, years = 5), 2)
```

year	saving	discount_factor	worth_today
1	0	0.95	0.00
2	816	0.91	740.14
3	816	0.86	704.89
4	816	0.82	671.33
5	816	0.78	639.36
6	816	0.75	608.91

Before I press Run: 816 AED saved in **year 4**.
At 5% a year, what is it worth in today's money? Guess a number.

About 670 AED: 816×0.82 . The **discount rate** says how fast future money loses value. Year 1 shows 0 because the saving starts a year after rollout.

The calculator

Discount rate (%)



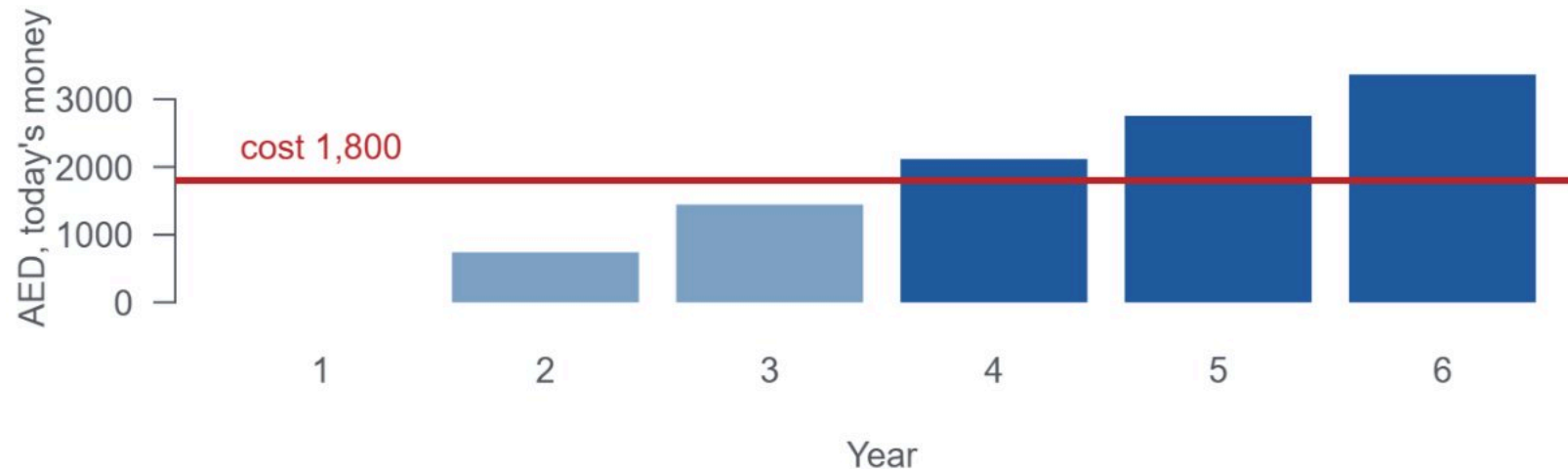
Years the saving lasts



Bars: savings so far, in today's money. It pays for itself once a bar reaches the red line.

`calculator(rate, years)`

5% discount, 5 years: savings worth 3,365 AED today. Ratio = $3,365 / 1,800 = 1.87$



Choice 1: the discount rate, on its own

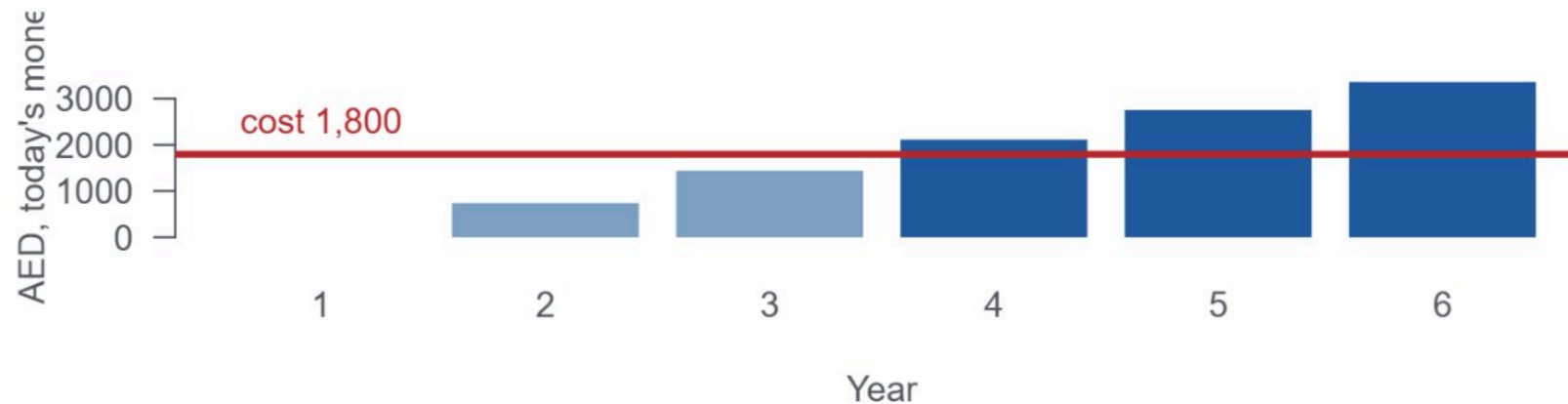
Discount rate (%)

5



`calculator(rate1, years = 5)`

5% discount, 5 years: savings worth 3,365 AED today. Ratio = $3,365 / 1,800 = 1.87$



Agencies typically use 3% to 8%. **At what rate does the ratio fall below 1?** Guess, then slide.

Around **24%**. Across 3% to 8% the ratio only moves from 2.02 to 1.68. The rate alone does not flip this decision.

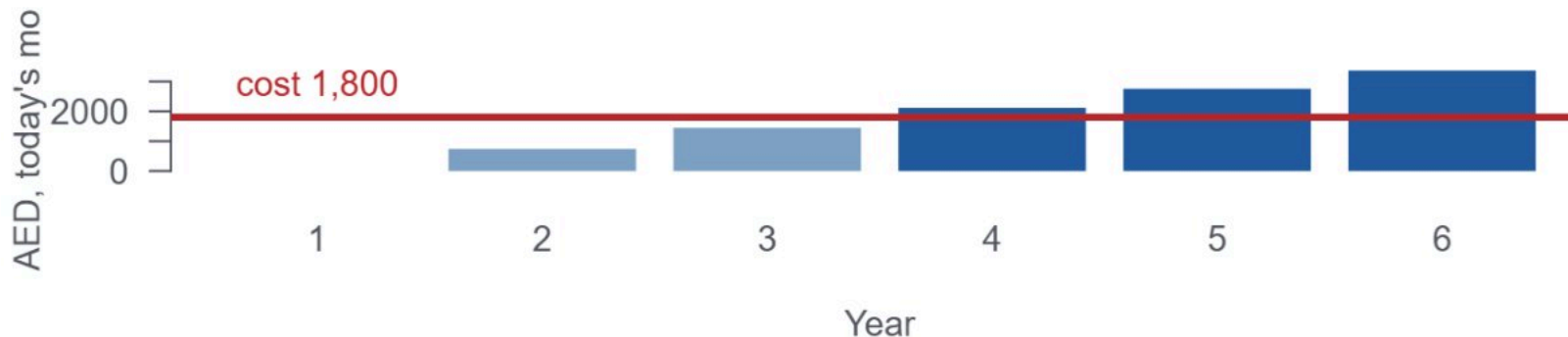
Choice 2: how long the saving lasts

Years the saving lasts (5% rate)



```
calculator(rate = 5, years = years2)
```

5% discount, 5 years: savings worth 3,365 AED today. Ratio = $3,365 / 1,800 = 1.87$



The Day 2 evaluations measured the saving for **one year**. Guess the ratio at two years, then slide down.

At two years the ratio is **0.80**: below break-even. Below about 2.5 years the programme does not pay for itself, and nothing in the evaluation measured how long the saving lasts.

Both choices together

Years of saving assumed	3%	5%	8%	12%	20%
2 years	0.84	0.80	0.75	0.68	0.58
3 years	1.25	1.18	1.08	0.97	0.80
5 years	2.02	1.87	1.68	1.46	1.13
8 years	3.09	2.79	2.41	2.01	1.45
10 years	3.76	3.34	2.82	2.29	1.58
	Discount rate				

Same programme, same cost, same measured saving in every cell; only the two choices differ.

Find the row that is below 1 whatever the rate.

Two years. If the saving lasts two years, no discount rate makes this worth doing. Read down the columns, not across the rows: the horizon matters far more than the rate.

3 · Three questions for any ratio

How to judge a cost-benefit claim

- 1 Are the benefits plausible?
- 2 Are all the costs included?
- 3 What would flip the result?

None of these needs a spreadsheet. They follow the pattern of the week: what is the number compared to, what is missing, and what would change the answer.

Question 1: are the benefits plausible?

R Code

↺ Start Over

▶ Run Code

```
1 low <- savings_table(5, 5, saving = 753) # the interval end closest to zero
2 high <- savings_table(5, 5, saving = 880)
3 round(data.frame(saving = c(753, 816, 880),
4                      ratio = c(ratio(low), ratio(savings_table(5, 5)), ratio(high))), 2)
```

saving	ratio
753	1.72
816	1.87
880	2.02

The saving here is **measured**. The three rows carry its confidence interval from Day 2 into the ratio.

The ratio stays between 1.72 and 2.02. The measured uncertainty barely moves it. The horizon, which nobody measured, moves it from 0.80 to 1.87.

Question 2: are all the costs included?

Costs left out, per business (AED)



```
tb <- savings_table(5, 5)
data.frame(cost_counted = 1800 + extra, ratio = round(ratio(tb, cost = 1800 + extra), 2))
```

cost_counted	ratio
1800	1.87

The 1,800 AED is the delivery cost only. Not counted: the authority's set-up and administration, the businesses' own time, anything the programme displaced.

How much would the missing costs have to add, per business, before the ratio falls below 1?

About **1,600 AED**. Every cost left out makes the ratio look better, so ask what is in the cost line.

Question 3: what would flip it? Your turn

Discount rate (%)	5	
Years the saving lasts	5	
Saving shrinks each year by (%)	0	
Costs left out (AED)	0	

```
tb <- savings_table(s_rate, s_years, decay = s_decay)
data.frame(ratio = round(ratio(tb, cost = 1800 + s_extra), 2),
            verdict = ifelse(ratio(tb, cost = 1800 + s_extra) >= 1, "pays for itself", "does not pay"))
```

ratio	verdict
1.87	pays for itself

Each group takes one card and tells me the settings. Is your scenario plausible for a waste-cost programme, and what is your verdict? Then: **how long do you think the saving lasts, and what evidence would you ask for?**

4 • Put AI to work

AI Snapshot: find the scenario that breaks it

Groups: paste the cost-benefit table and this prompt.

“Generate one plausible scenario in which this programme’s benefit-cost ratio falls below 1. Then tell me whether that scenario is realistic.”

Check each claim against the calculator. Which is right, which is wrong, and which is true but no use to you? Four minutes.

A response like this is possible, especially with a short or leading prompt:

The ratio falls below 1 if the discount rate is raised. **At a discount rate of around 40%, the ratio drops below break-even**, which would mean the programme does not pay for itself and should not be funded.

Alternatively, if the programme’s benefits are delayed by a few years, the ratio also falls below 1. Both scenarios are **realistic and commonly observed in infrastructure appraisal.**

Debrief

The arithmetic holds. At 40% the ratio is below 1; in fact it crosses at about 24%.

1 • It picked the easiest lever. No agency uses a rate that high. The model answered “name a scenario”, not “name the one that matters”.

2 • It came near the right lever and gave no number. “Benefits delayed” is close to the horizon, which decides this programme, but it never says how many years or what the ratio becomes.

3 • “Commonly observed in infrastructure appraisal.” Unsourced, and about a different kind of programme.

The model invented no number and misread no table. It chose what was easy to compute. Ask for the assumption that matters most, and ask for the number.

The same question, a better prompt

What people actually type

Generate one plausible scenario in which this benefit-cost ratio falls below 1.

“Plausible” is not defined, so the model may pick whatever is easiest to compute.

What to type instead

Here is a cost-benefit model. The cost and annual saving are measured; the discount rate and the number of benefit years are assumptions. Tell me which **single assumption** moves the ratio most, give the ratio under a realistic alternative, and say whether a government agency would normally make that assumption. **Ask me questions before you answer.**

Run both. Does the second rank the assumptions before choosing one? What did each explain well?

5 • Close

Two tests, one programme

Day 2 rule cut costs by 1,000 AED in a year
What was found 816 AED, real and short of the bar
Verdict do not scale on that evidence

Today's ratio 1.87 AED back per AED spent
What it assumes the saving lasts five years
Verdict worth funding, if that holds

Same programme, estimate and cost. The rule asks whether the effect is big enough to scale; the ratio asks whether the spending pays for itself over time. The rule rests on a measured saving. The ratio also rests on how long that saving lasts, which nobody measured.

Which would you put in front of the minister, and what would you add to it?

What today gives you

A ratio is a model	it contains choices as well as evidence
The rate	3% to 8% barely moves this one
The horizon	decides it, and the evaluation did not measure it
The costs	anything left out flatters the ratio

The sentence to take to a meeting: **“What assumption would have to change for this ratio to fall below 1, and what is your evidence for it?”**

Where this sits

- Day 1** is the number answering your question, and is it big enough?
- Day 2** which method produced it, and what does that method assume?
- Today** what is it worth, and which assumption decides that?
- Next session** reading the charts that carry these claims into a meeting